3.4.4 Risk-based boundaries, attribute lifetimes, and re-proofing

The attributes associated with an account change over time. For those determined by the system itself, they can be updated as they change. But others, provided by the enrollee, might change without the system's knowledge or action. Some attributes, formally known as *traits*, are unlikely to change—these include birthdate and eye color. But others, like marital status or role inside another organization (think about my relationship to the supplier account from the last section), often change without any signal that the identity system can use to trigger an update.

Because of this, many IAL2 and IAL3 systems require reproofing at regular intervals. Proofed attributes, like those from a driver's license or state ID, can record information about the underlying document's expiration date. This can trigger a need for re-proofing, but that depends on the organization's policies and regulatory environment. The birthdate on your driver's license is still the same even if the license has expired. In other cases, some event such as a change in relationship status, role change, detection of fraud, or change in behavior by an account holder may require reproofing.

3.4.5 The role of credentials and authentication factors

One of the primary purposes of enrollment is to provide the enrollee with authentication credentials they can use to prove who they are when they come back to the system. Some of the attributes gathered during enrollment, like email addresses and mobile phone numbers, are important to get right, so the system often performs automated proofing to verify these. If the system uses digital certificates or biometrics, these are also recorded during enrollment. Finally, in both self-service and administrative enrollments, the system usually assigns initial roles and sets defaults that are helpful later when the enrollee uses the system.

3.5 Authentication factors

Authentication is how we recognize the rightful account holder, even when they aren't physically present. In face-to-face interactions, identity